

Traditional

2007
DNN

Class.

→ □

- 1) Radar.
- 2) Lidar.
- 3) Magnetic
- 4) CT's.
- 5) Ultrasound.

Session #12:

Semantic Segmentation and CV Metrics

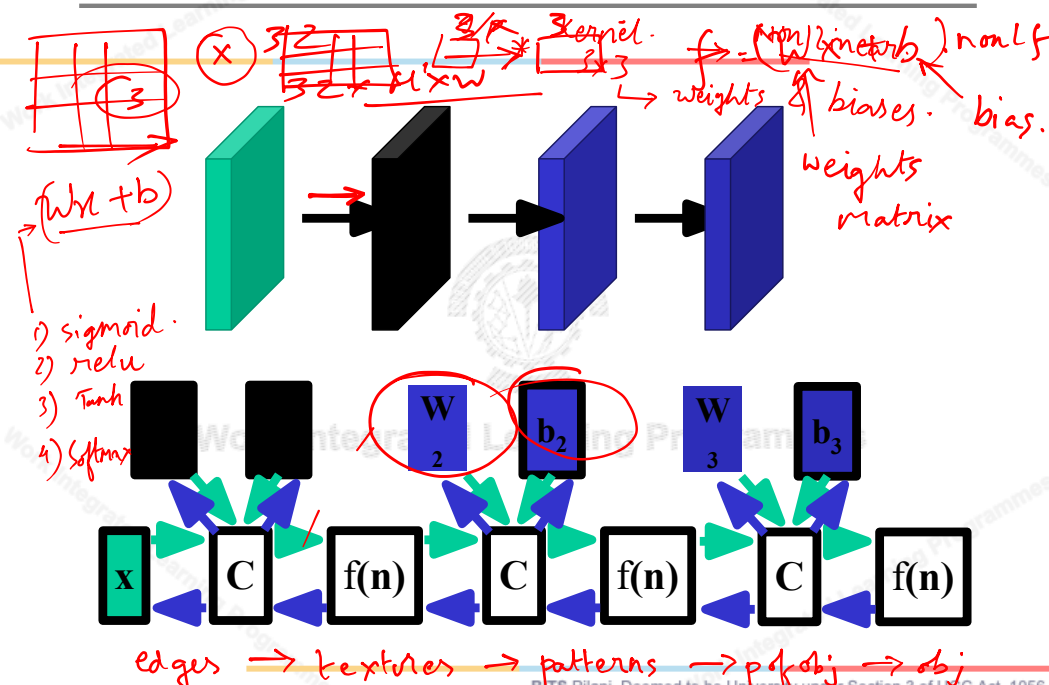
Dhruba Adhikary

Dhruba.a@wilp.bits-pilani.ac.in

Intensity.
visible spectrum
KDE
3D
Vision.



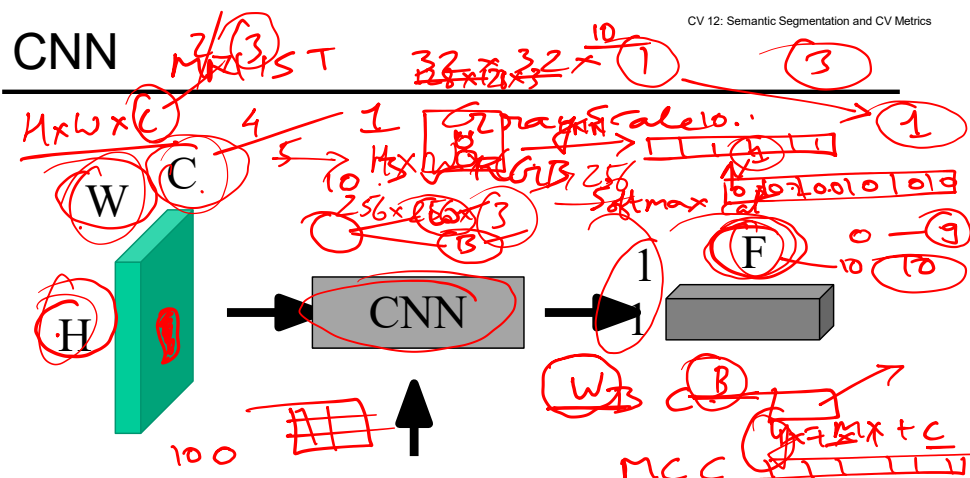
Convolutional Neural Network (CNN)



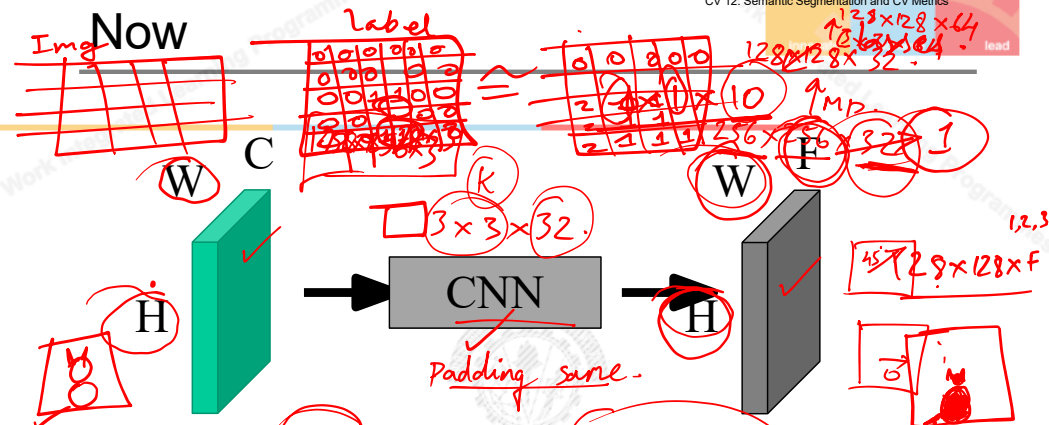
Acknowledgement: Slide Materials adopted from EECS 442 – David Fouhey
Winter 2023, University of Michigan

CV 12: Semantic Segmentation and CV Metrics

CNN



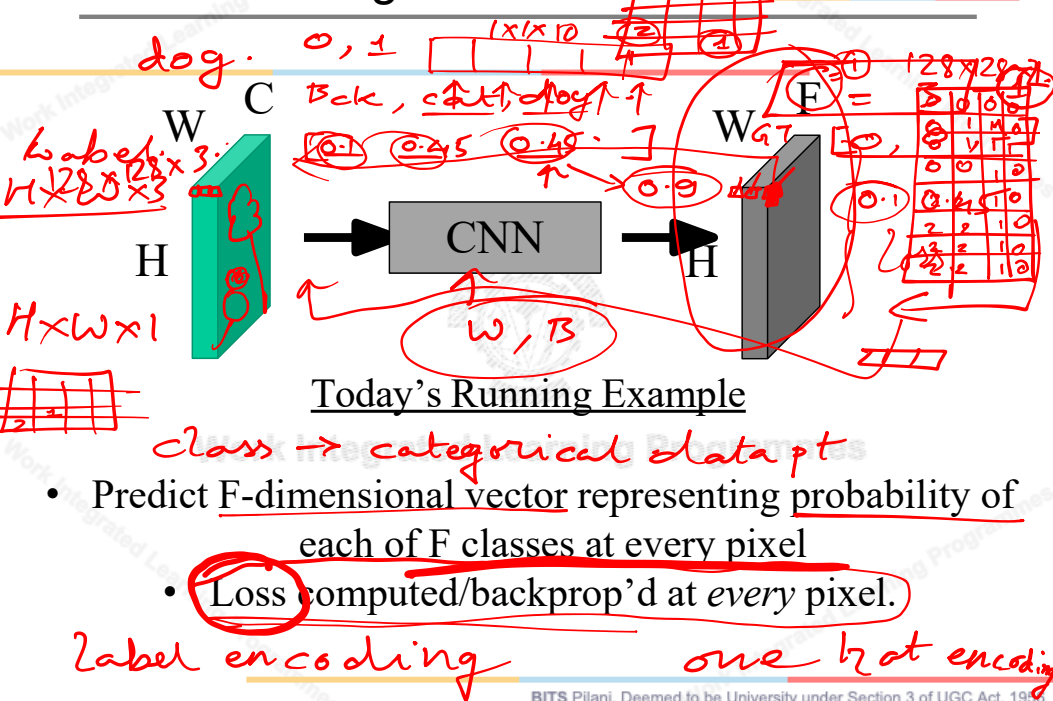
CV 12: Semantic Segmentation and CV Metrics



Convert $H \times W$ image into a F -dimensional vector

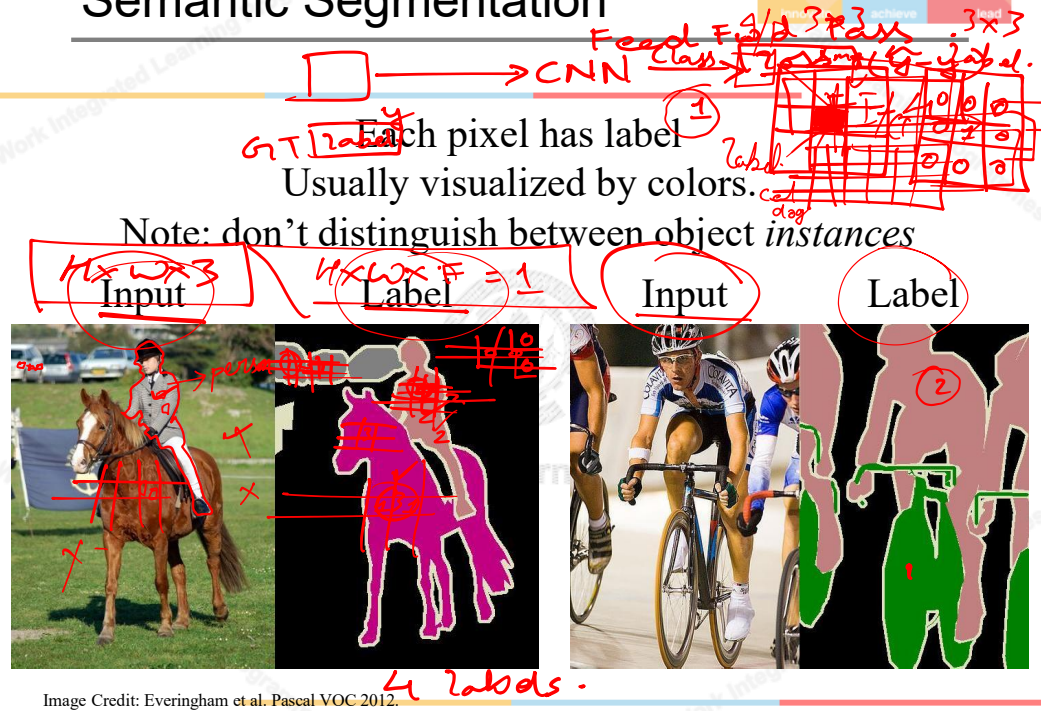
- Which pixels in this image are a cat?
- How far is each pixel away from the camera?
- Which pixels of this image are fake?

Semantic Segmentation



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Semantic Segmentation



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Semantic Segmentation

CVAT

“Semantic”: a usually meaningless word and an indication that someone is trying to trick you.

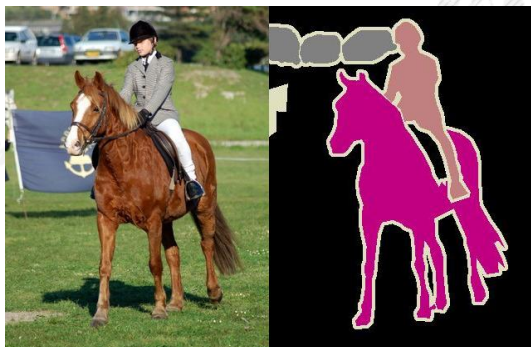
Meant to indicate here that we're **naming** things.

Input

Label

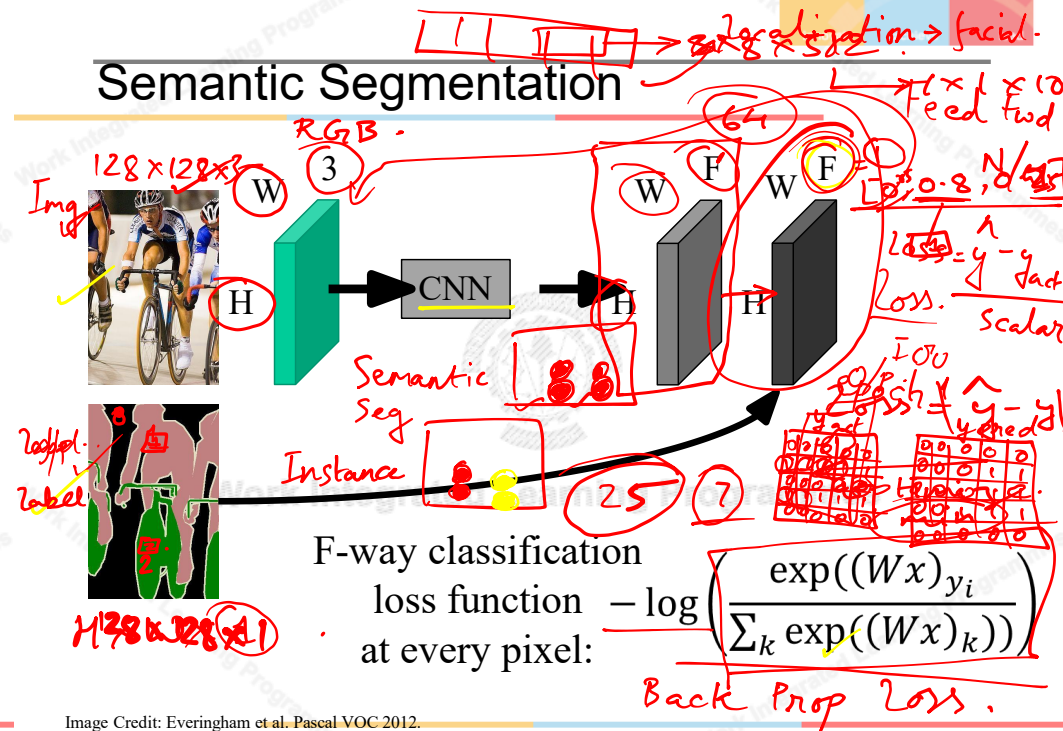
Input

Label



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Semantic Segmentation



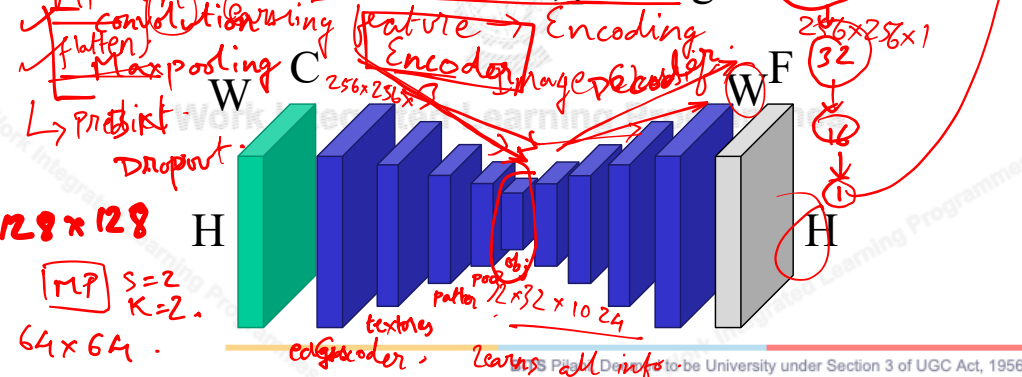
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Encoder-Decoder

Key idea: First downsample towards middle of network. Then upsample from middle.

How do we downsample?

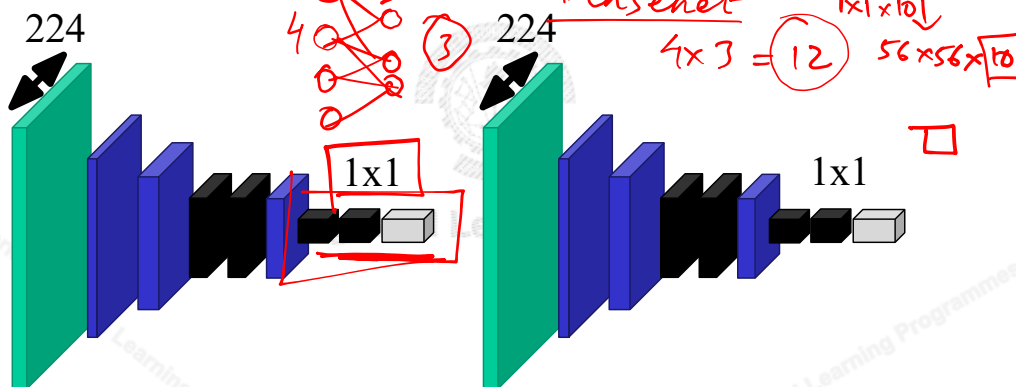
Convolutions, pooling



Where Do We Get Parameters?

Convnet that maps
images to vectors

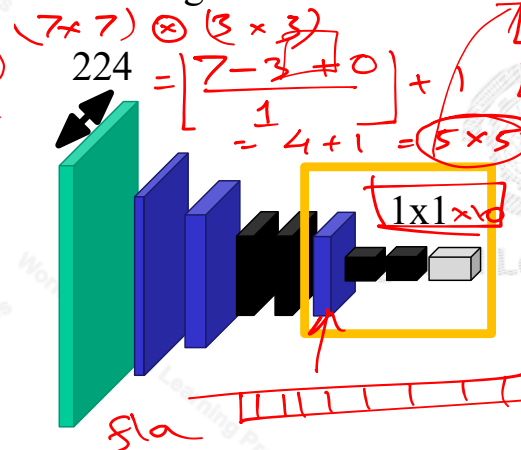
Convnet that maps images to images



What if we make the input bigger?

Where Do We Get Parameters?

Convnet that maps
images to vectors



Recall that we can rewrite
any vector-vector
operations via 1x1
convolutions

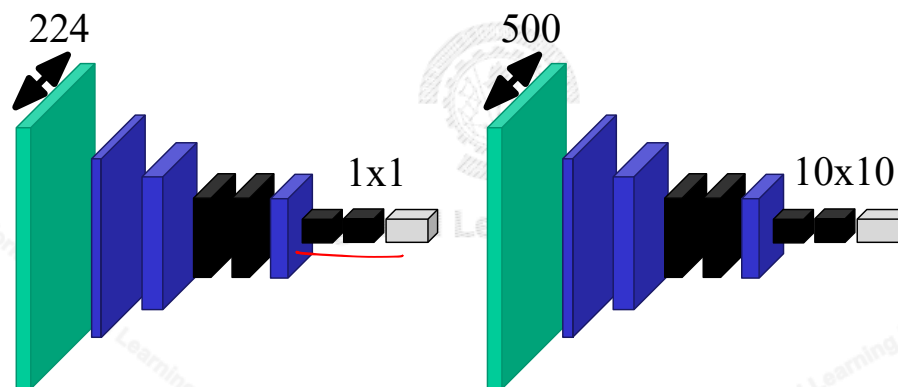
$$32 \times 32 \times \boxed{64} =$$

Long et al. *Fully Convolutional Networks For Semantic Segmentation*. CVPR 2014

Where Do We Get Parameters?

Convnet that maps images to vectors

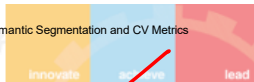
Convnet that maps images to images



Since it's convolution, can reuse an image network

semantic Segmentation and CV Metrics

innovate achieve lead

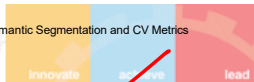


semantic Segmentation and CV Metrics

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semantic Segmentation and CV Metrics

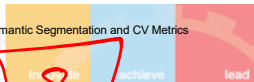
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semantic Segmentation and CV Metrics



semantic Segmentation and CV Metrics

semantic Segmentation and CV Metrics



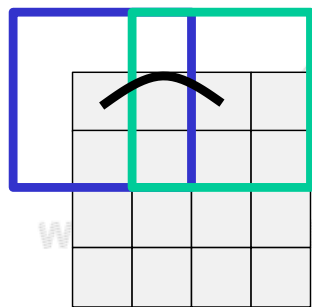
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semantic Segmentation and CV Metrics

Recall: Convolution

5x5 $\rightarrow$ 3x3

3x3 Convolution, Stride 2, Pad 1



MSD.
Mobile net



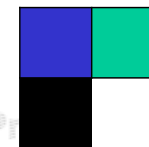
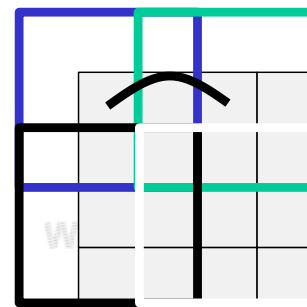
Dot product
between filter f
and input

Example Credit: L. Fei-Fei, J. Johnson, S. Yeung

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Recall: Convolution

3x3 Convolution, Stride 2, Pad 1



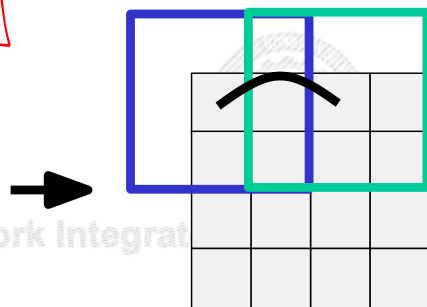
Example Credit: L. Fei-Fei, J. Johnson, S. Yeung

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Transpose Convolution

3x3 Transpose Convolution, Stride 2, Pad 1

3x3.



Output is filter F
weighted by input
 I

$I_{11}F$ $I_{12}F$

5x5 g/p

7x7 O/P.

Example Credit: L. Fei-Fei, J. Johnson, S. Yeung

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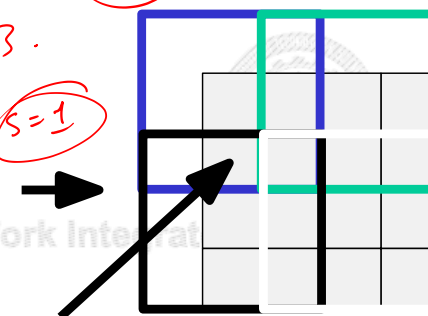
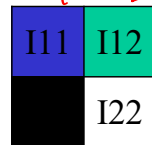
Transpose Convolution

3x3 Transpose Convolution, Stride 2, Pad 1

O/P = $(I/P - 1) \times S + K - 2P$.

7 = $(5 - 1) \times 2 + 3 - 0$.

7 = 4s + 3.
4 = 4s



Output is filter F
weighted by input
 I

$I_{11}F$ $I_{12}F$

$I_{21}F$

Sum outputs at overlap (e.g.,
from $I_{11}F$ and $I_{21}F$)

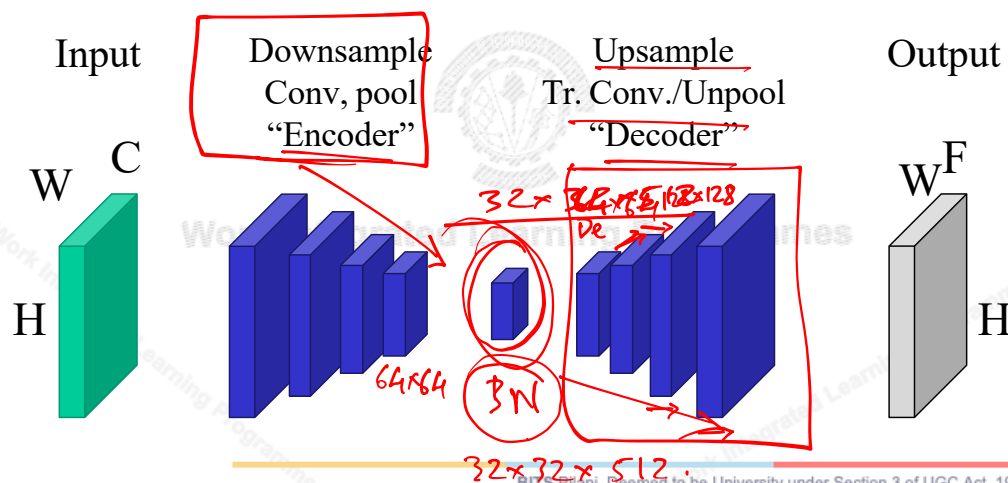
11 = 4s + 3
11 - 3 = 4s
s = 8/4 = 2

Example Credit: L. Fei-Fei, J. Johnson, S. Yeung

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Putting it Together

Convolutions + pooling downsample/compress/encode
 Transpose convs./unpoolings upsample/uncompress/decode

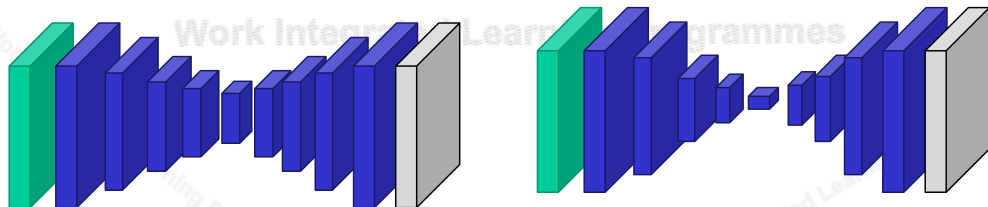


Putting It Together – Block Sizes

- Networks come in lots of forms
- **Don't take any block sizes literally.**
- Often (not always) keep some spatial resolution

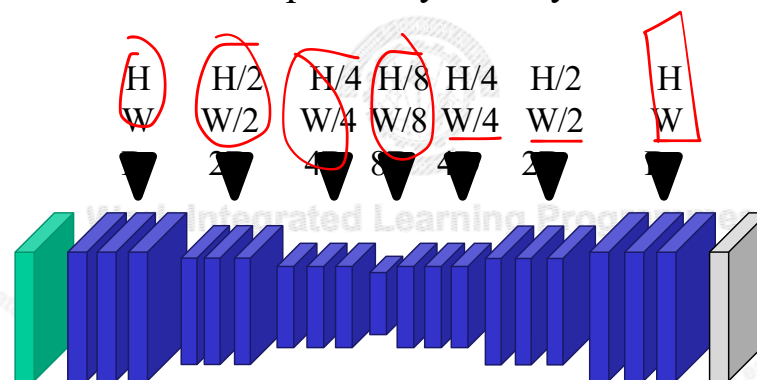
Encode to spatially smaller tensor, then decode.

Encode to 1D vector then decode



Putting It Together – Block Sizes

- Often multiple layers at each spatial resolution.
- Often halve spatial resolution and double feature depth every few layers



Missing Details

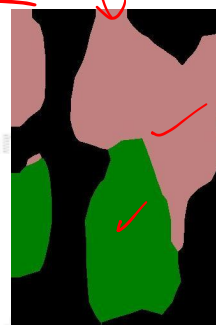
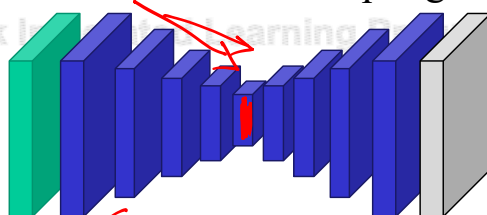


While the output is HxW, just upsampling often produces results without details/not aligned with the image.

Why?



Information about details lost when downsampling!



Result from Long et al. *Fully Convolutional Networks For Semantic Segmentation*. CVPR 2014

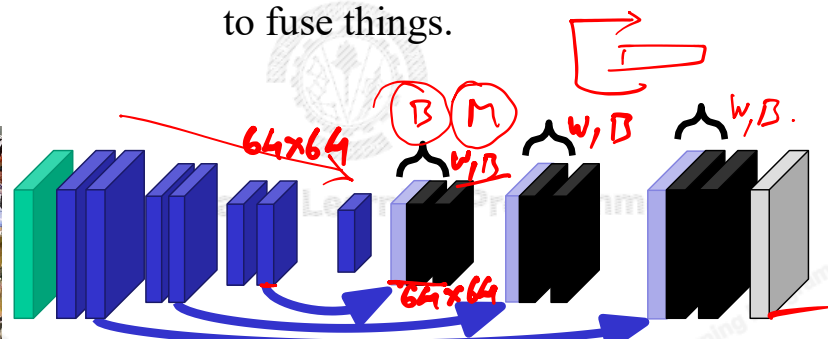
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Missing Details

How do you send details forward in the network?

You copy the activations forward.

Subsequent layers at the same resolution figure out how to fuse things.



Copy

Concat.

Result from Long et al. *Fully Convolutional Networks For Semantic Segmentation*. CVPR 2014

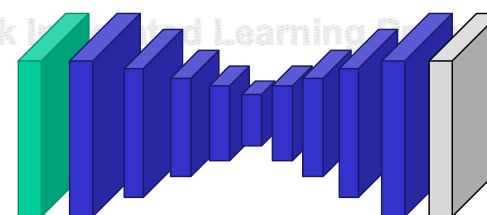
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Missing Details

Where is the useful information about the high-frequency details of the image?



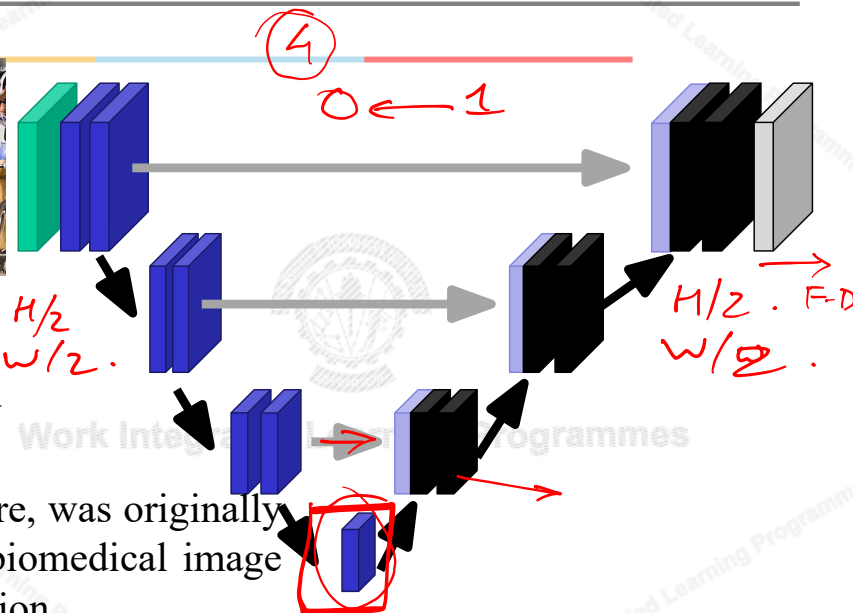
A B C D E



Result from Long et al. *Fully Convolutional Networks For Semantic Segmentation*. CVPR 2014

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U-Net

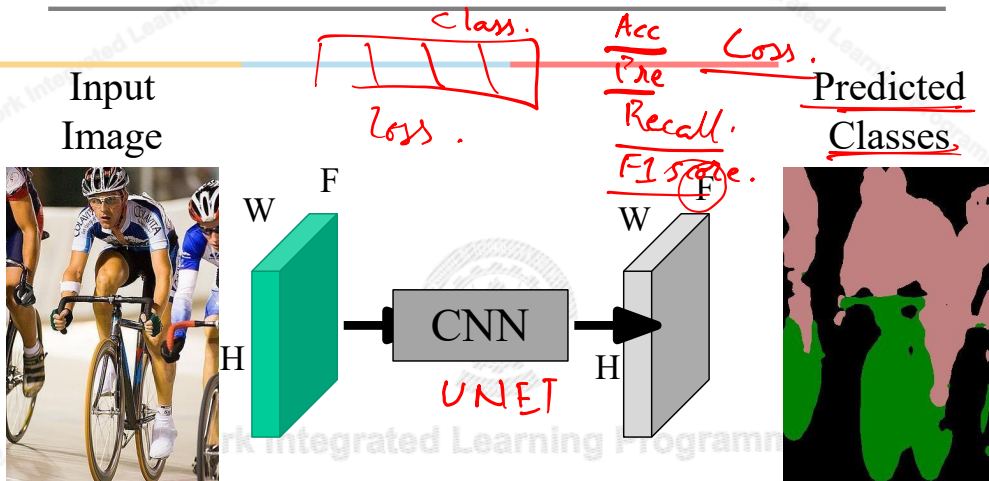


Extremely popular architecture, was originally used for biomedical image segmentation.

Ronneberger et al. "U-Net: Convolutional Networks for Biomedical Image Segmentation". MICCAI 2015

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Evaluating Pixel Labels



How do we convert final HxWxF into labels?

argmax over labels

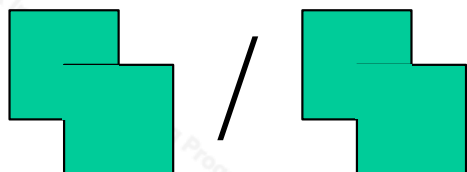
Evaluating Semantic Segmentation

Jou

Prediction and ground-truth are images where each pixel is one of F classes.

Accuracy: $\text{mean}(\hat{y} = y)$

Intersection over union, averaged over classes



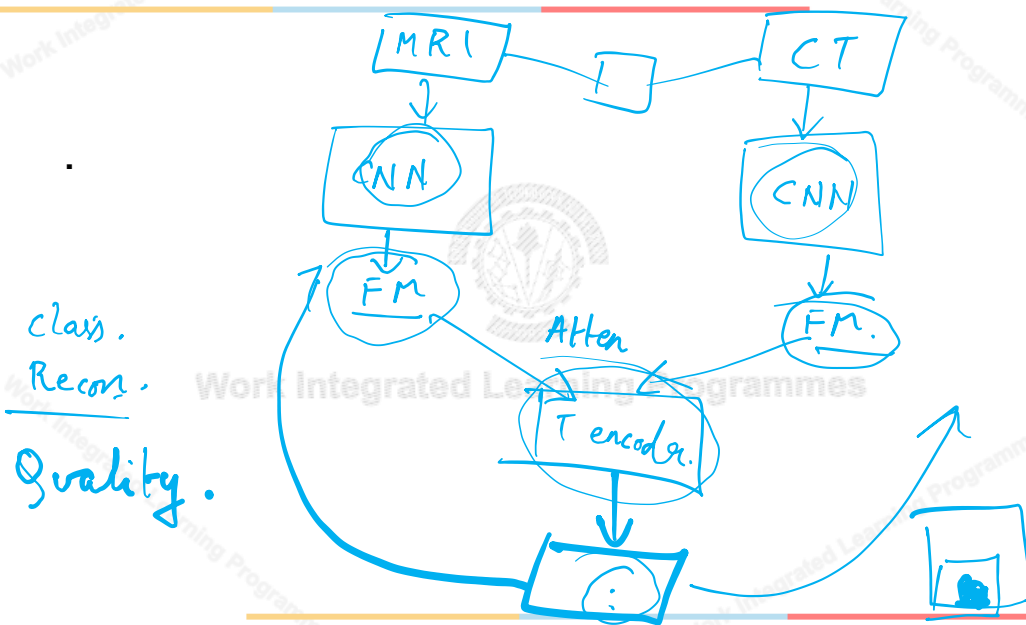
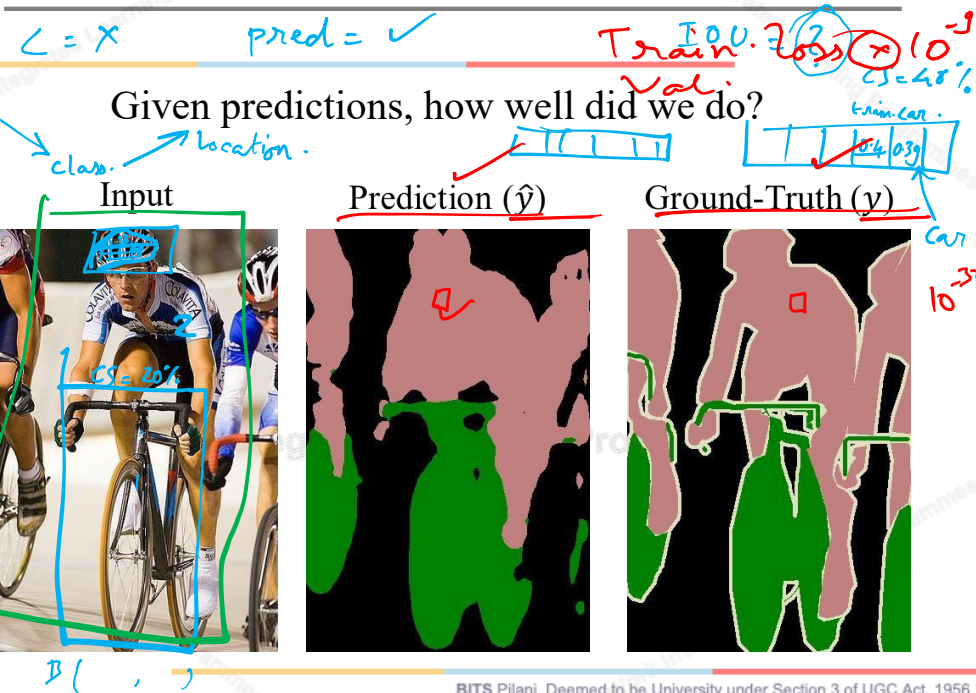
Prediction ($\hat{y}$)



Ground-Truth (y)



Evaluating Semantic Segmentation



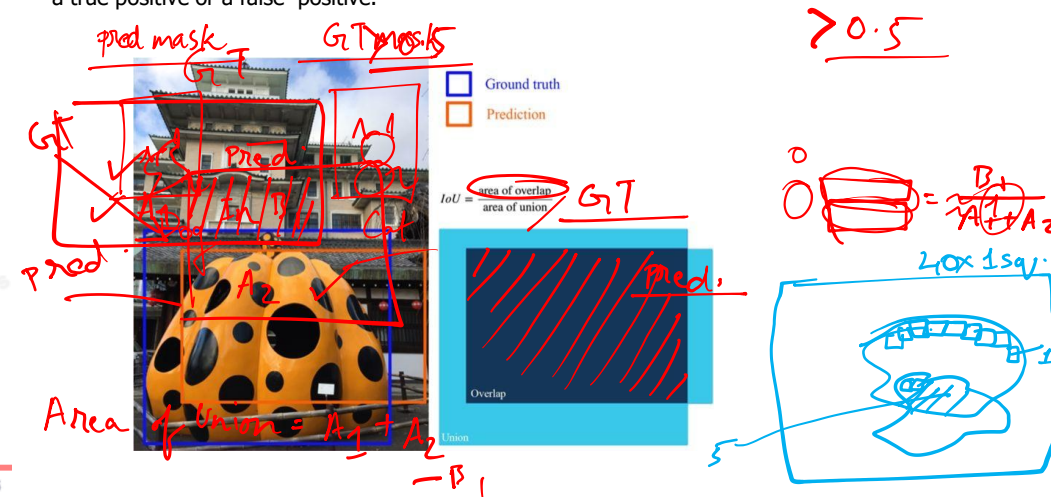
Performance Metrics for Segmentation

- see materials

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IoU (Intersection over union)

- IoU measures the overlap between two boundaries.
- We use that to measure how much our predicted boundary overlaps with the ground truth (the real object boundary).
- In some datasets, we predefine an IoU threshold (say 0.5) in classifying whether the prediction is a true positive or a false positive.



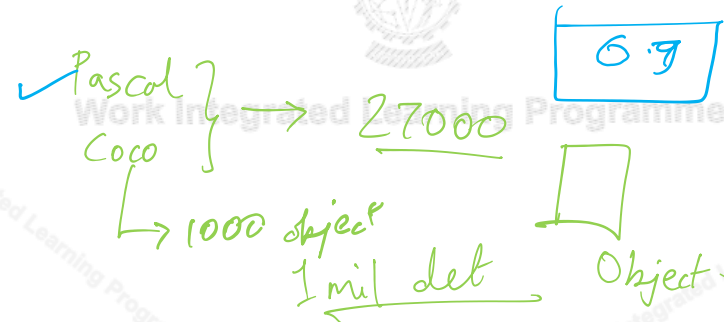
mAP for Object Detection

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mAP for Object Detection

Source: Karel Hora
 Brno University of Technology / Czech Technical University in Prague
horak@feec.vutbr.cz

- mAP stands for mean Average Precision.
- AP is a popular metric in measuring the accuracy of object detectors like Faster R-CNN, SSD, etc.
- Average precision computes the average precision value for recall value over 0 to 1. It sounds complicated but actually pretty simple as we illustrate it with an example.
- But before that, we will do a quick recap on precision, recall, and IoU first.



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Precision & recall

- Precision** measures how accurate is your predictions. i.e. the percentage of your predictions are correct.
- Recall** measures how good you find all the positives. For example, we can find 80% of the possible positive cases in our top K predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

TP = True positive

TN = True negative

FP = False positive

FN = False negative

- F1 score** is a measure combining Precision and Recall.

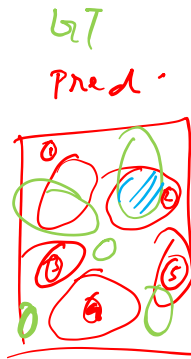
$$F1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

$$A = 0.6$$

$$B = 0.7$$

$$C = 0$$

$$D = 0.25$$



IOU =

Numerical IOU

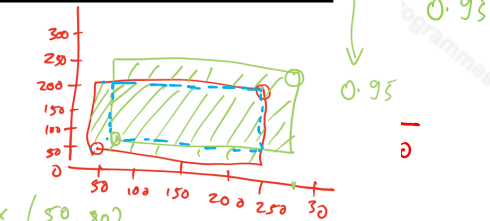
$$\text{Dice Coeff} = 2 \cdot \text{IOU} = \frac{21600}{37800} = 0.57$$

Numerical

$$\text{Dice Coeff} = 2 \cdot \text{IOU} = \frac{21600}{37800} = 0.57$$

IOU? GT x_1, y_1, x_2, y_2
 Pred x_1, y_1, x_2, y_2

Intersection Area = $\max(70, 50), \max(50, 30)$
 $= (70, 30)$
 $= \min(250, 280), \min(200, 220)$
 $= (250, 200)$
 Int ht = $250 - 70 = 180$
 Int wd = $200 - 30 = 170$
 Area = $180 \times 170 = 30600$



$$L = 250 - 50 = 200$$

$$B = 150$$

Union Area = $(200 - 50) \times (250 - 50) + (280 - 70) \times (220 - 30)$
 $= 21600$
 $= 37800$

C Act, 1956

Average Precision

Top 5 pred.



mAP 0.75 mAP 0.65

- Let us create an over-simplified example in demonstrating the calculation of the average precision.
- In this example, the whole dataset contains five apples only. We collect all the predictions made for apples in all the images and rank it in descending order according to the predicted confidence level. The second column indicates whether the prediction is correct or not. In this example, the prediction is correct if IoU ≥ 0.5 .

Rank	IoU	Correct?	Precision	Recall
1	0.99	True	1.0	0.2
2	0.95	True	1.0	0.4
3	0.94	False	0.67	0.4
4	0.6	False	0.5	0.4
5	0.6	False	0.4	0.4
6	0.7	True	0.57	0.6
7		True	0.5	0.8
8		False	0.5	0.8
9		False	0.44	0.8
10		True	0.5	1.0

Average Precision

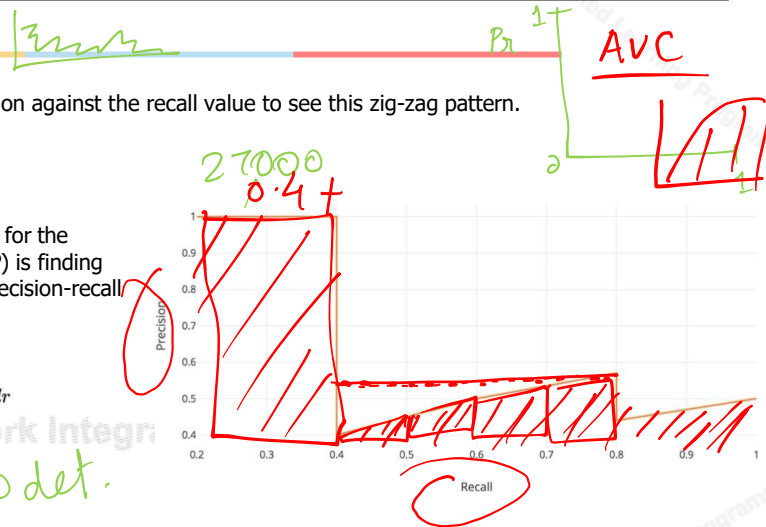
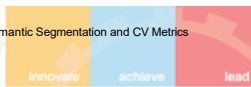


- Let us take the row with rank #3 and demonstrate how precision and recall are calculated first.
- Precision** is the proportion of TP = $2/3 = 0.67$.
- Recall** is the proportion of TP out of the possible positives = $2/5 = 0.4$.
- Recall values increase as we go down the prediction ranking. However, precision has a zigzag pattern — it goes down with false positives and goes up again with true positives.

Rank	Correct?	Precision	Recall
1	True	1.0	0.2
2	True	1.0	0.4
3	False	0.67	0.4
4	False	0.5	0.4
5	False	0.4	0.4
6	True	0.57	0.6
7	True	0.57	0.8

Average Precision

CV 12: Semantic Segmentation and CV Metrics



- Let us plot the precision against the recall value to see this zig-zag pattern.
- The general definition for the Average Precision (AP) is finding the area under the precision-recall curve.

$$AP = \int_0^1 p(r) dr$$

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Average Precision

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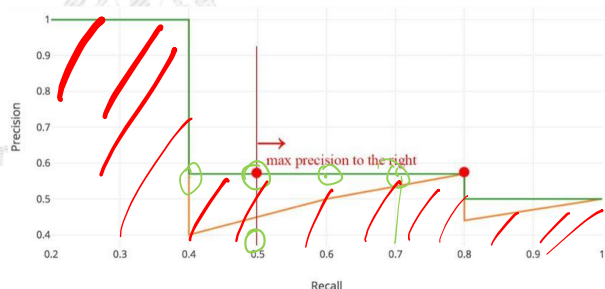
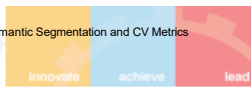


- Precision and recall are always between 0 and 1. Therefore, AP falls within 0 and 1 also.
- Before calculating AP for the object detection, we often smooth out the zigzag pattern first.
- Graphically, at each recall level, we replace each precision value with the maximum precision value to the right of that recall level.

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Average Precision

CV 12: Semantic Segmentation and CV Metrics



- So the orange line is transformed into the green line and the curve will decrease monotonically instead of the zigzag pattern.
- The calculated AP value will be less susceptible to small variations in the ranking.
- Mathematically, we replace the precision value for recall $\hat{r}$ with the maximum precision for any recall $\geq \hat{r}$.

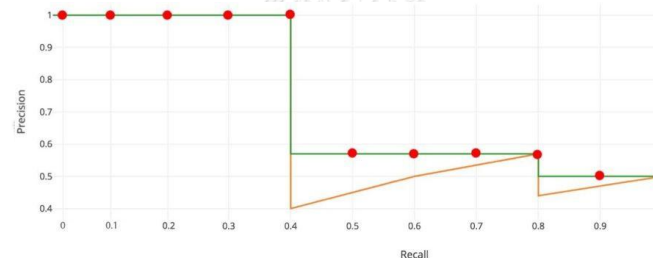
$$p_{interp}(r) = \max_{\tilde{r} \geq r} p(\tilde{r})$$

0.7
0.6

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Interpolated AP

CV 12: Semantic Segmentation and CV Metrics



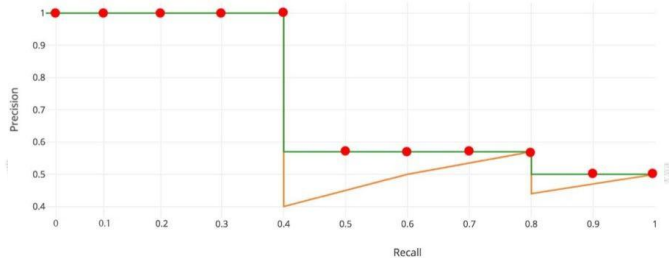
- PASCAL VOC** is a popular dataset for object detection.
- For the PASCAL VOC challenge, a prediction is positive if IoU ≥ 0.5 . Also, if multiple detections of the same object are detected, it counts the first one as a positive while the rest as negatives.
- In Pascal VOC2008, an average for the 11-point interpolated AP is calculated.

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Interpolated AP

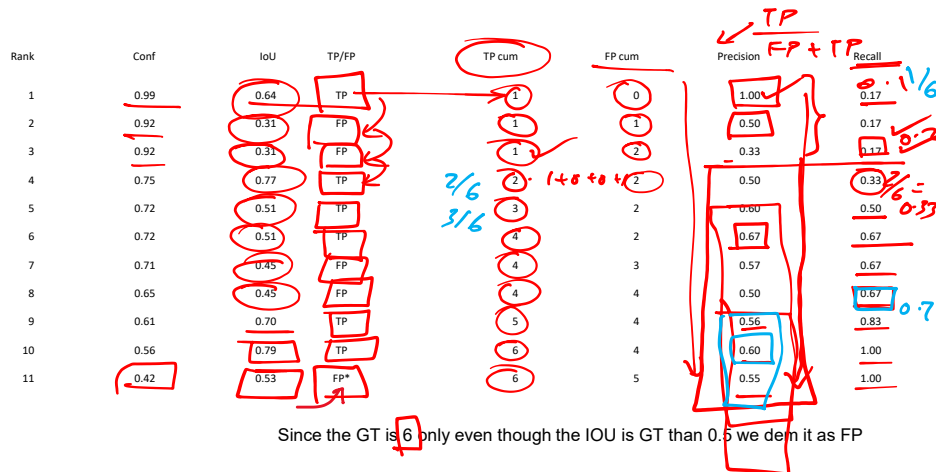
- First, we divide the recall value from 0 to 1.0 into 11 points — 0, 0.1, 0.2, ..., 0.9 and 1.0.
- Next, we compute the average of maximum precision value for these 11 recall values.

$$AP = \frac{1}{11} \times (AP_r(0) + AP_r(0.1) + \dots + AP_r(1.0))$$

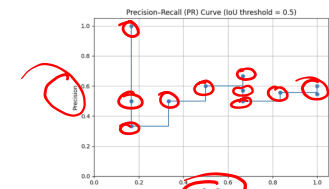


- In our example, $AP = (5 \times 1.0 + 4 \times 0.57 + 2 \times 0.5)/11$

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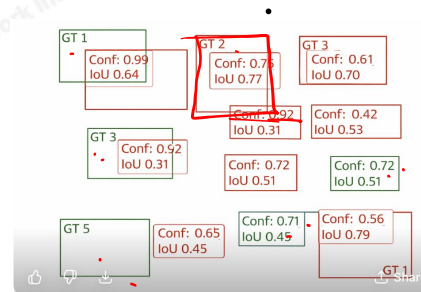
FP at rank 11 because we already reached **6 TPs** (all GT matched). In normal matching, any additional detections are **duplicates** → FP.



Here's the Precision-Recall (PR) curve (using the 11 detections visible in your image, IoU threshold = 0.5):

$$Recal = \frac{TP}{TP + FN}$$

5 GT objects of interest



Assumptions used (standard object detection evaluation)

- #GT = 6
- IoU threshold = 0.5
- Detections are processed in descending confidence
- A detection counts as TP if $IoU \geq 0.5$ and we still have an unmatched GT available (so TP cannot exceed 6; extra "good IoU" detections become FP due to duplicates)

Below are the interpolated-precision tables for 9-point and 11-point evaluation, using your PR points (6 GT, $IoU \geq 0.5$) from the previous step.

How interpolated precision is taken at a recall level r

For a chosen recall level r , VOC-style interpolation uses:

$$p_{\text{interp}}(r) = \max_{r' \geq r} p(r')$$

Meaning: look at all PR points whose recall is $\geq r$, and take the maximum precision among them.

Example (from your PR points):

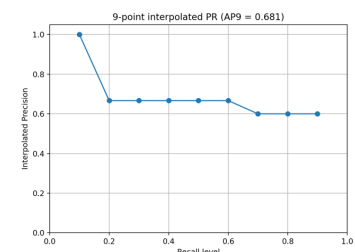
At $r = 0.7$, eligible recalls are 0.8333, 1.0, 1.0 with precisions 0.5556, 0.6, 0.5455.

So:

$$p_{\text{interp}}(0.7) = \max(0.5556, 0.6, 0.5455) = 0.6$$

9-point interpolation table (recall levels 0.1 to 0.9)

Recall level (r)	$p_{\text{interp}}(r)$
0.1	1.000
0.2	0.667
0.3	0.667
0.4	0.667
0.5	0.667
0.6	0.667
0.7	0.600
0.8	0.600
0.9	0.600



Formula

$$AP_9 = \frac{1}{9} \sum_{r \in \{0.1, \dots, 0.9\}} p_{\text{interp}}(r)$$

1 class.

Sum:

$$1 + 3.333333 + 1.8 = 6.133333 \quad 10 \text{ class}$$

Divide by 9:

$$AP_9 = \frac{6.133333}{9} = 0.681481$$

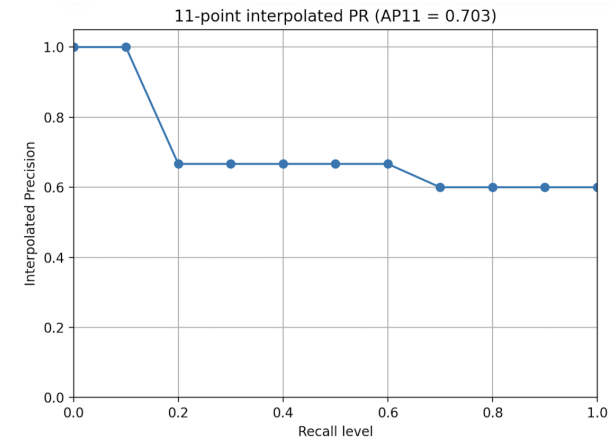
✓ 9-point mAP (AP₉) = 0.6815 (≈ 0.681)

mAP: $\frac{\text{car} \cdot 0.68 + \text{bike} \cdot 0.5 + \text{human} \cdot 0.9 + \text{cat} \cdot 0.2}{4}$

11-point interpolation table (recall levels 0.0 to 1.0)

Recall level (r)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0

(p _{interp})(r)
1.000
1.000
0.667
0.667
0.667
0.667
0.667
0.600
0.600
0.600
0.600



Formula

$$AP_{11} = \frac{1}{11} \sum_{r \in \{0.0, 0.1, \dots, 1.0\}} p_{\text{interp}}(r)$$

Substitute values

$$AP_{11} = \frac{1}{11} \left(2 + 5 \cdot \frac{2}{3} + 4 \cdot 0.6 \right)$$

Sum:

$$2 + 3.333333 + 2.4 = 7.733333$$

Divide by 11:

$$AP_{11} = \frac{7.733333}{11} = 0.703030$$

✓ 11-point mAP (AP₁₁) = 0.7030 (≈ 0.703)

Interpolated AP

CV 12: Semantic Segmentation and CV Metrics



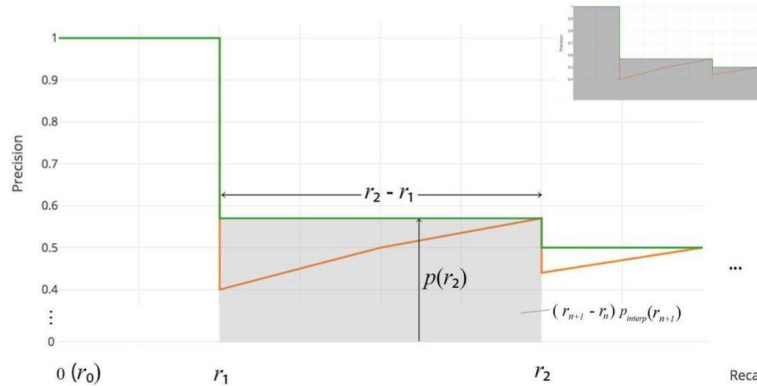
mAP 0.5 0.7

IOU

- When AP_r turns extremely small, we can assume the remaining terms to be zero. i.e. we do not necessarily make predictions until the recall reaches 100%.
- If the possible maximum precision levels drop to a negligible level, we can stop.
- For 20 different classes in PASCAL VOC, we compute an AP for every class and also provide an average for those 20 AP results.
- However, this interpolated method is an approximation which suffers two issues:
 1. It is less precise.
 2. Second, it lost the capability in measuring the difference for methods with low AP.
- Therefore, a different AP calculation is adopted after 2008 for PASCAL VOC.

AP (Area under curve AUC)

- For later Pascal VOC competitions, VOC2010–2012 samples the curve at all unique recall values ($r_1, r_2, \dots$), whenever the maximum precision value drops.
- With this change, we are measuring the exact area under the precision-recall curve after the zigzags are removed.

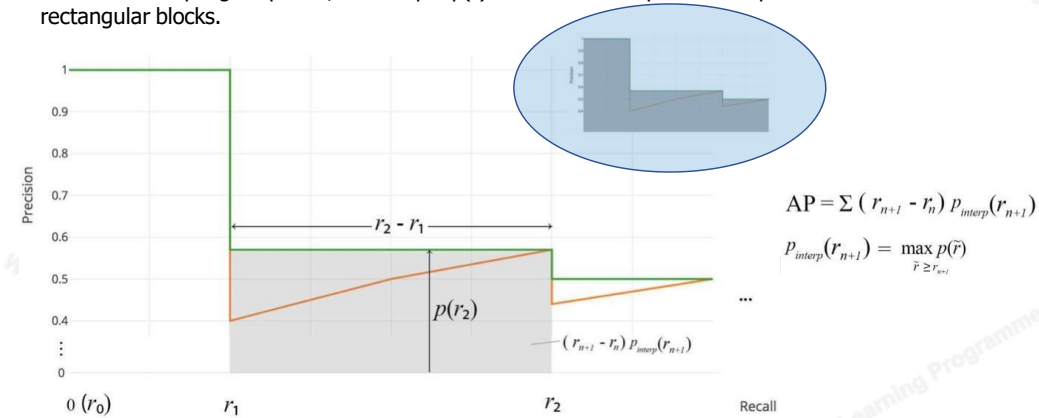


COCO mAP

- Latest research papers tend to give results for the COCO dataset only. In COCO mAP, a 101-point interpolated AP definition is used in the calculation.
- For COCO, AP is the average over multiple IoU (the minimum IoU to consider a positive match).
- AP@[.5:.95]** corresponds to the average AP for IoU from 0.5 to 0.95 with a step size of 0.05.
- For the COCO competition, AP is the average over 10 IoU levels on 80 categories (AP@[.50:.05:.95]: start from 0.5 to 0.95 with a step size of 0.05).

AP (Area under curve AUC)

- No approximation or interpolation is needed.
- Instead of sampling 11 points, we sample $p(r_i)$ whenever it drops and computes AP as the sum of the rectangular blocks.



COCO mAP

- The following are some other metrics collected for the COCO dataset:

Average Precision (AP):	
AP	% AP at IoU=.50:.05:.95 (primary challenge metric)
AP _{IoU=.50}	% AP at IoU=.50 (PASCAL VOC metric)
AP _{IoU=.75}	% AP at IoU=.75 (strict metric)
AP Across Scales:	
AP _{small}	% AP for small objects: area < 32 ²
AP _{medium}	% AP for medium objects: 32 ² < area < 96 ²
AP _{large}	% AP for large objects: area > 96 ²
Average Recall (AR):	
AR ^{max=1}	% AR given 1 detection per image
AR ^{max=10}	% AR given 10 detections per image
AR ^{max=100}	% AR given 100 detections per image
AR Across Scales:	
AR _{small}	% AR for small objects: area < 32 ²
AR _{medium}	% AR for medium objects: 32 ² < area < 96 ²
AR _{large}	% AR for large objects: area > 96 ²

COCO mAP

- Example: the AP result for the YOLOv3 detector:

	backbone	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
<i>Two-stage methods</i>							
Faster R-CNN++ [3]	ResNet-101-C4	34.9	55.7	37.4	15.6	38.7	50.9
Faster R-CNN w FPN [6]	ResNet-101-FPN	36.2	59.1	39.0	18.2	39.0	48.2
Faster R-CNN by G-RMI [4]	Inception-ResNet-v2 [19]	34.7	55.5	36.7	13.5	38.1	52.0
Faster R-CNN w TDM [18]	Inception-ResNet-v2-TDM	36.8	57.7	39.2	16.2	39.8	52.1
<i>One-stage methods</i>							
YOLOv2 [13]	DarkNet-19 [13]	21.6	44.0	19.2	5.0	22.4	35.5
SSD513 [9, 2]	ResNet-101-SSD	31.2	50.4	33.3	10.2	34.5	49.8
DSSD513 [2]	ResNet-101-DSSD	33.2	53.3	35.2	13.0	35.4	51.1
RetinaNet [7]	ResNet-101-FPN	39.1	59.1	42.3	21.8	42.7	50.2
RetinaNet [7]	ResNeXt-101-FPN	40.8	61.1	44.1	24.1	44.2	51.2
YOLOv3 608 × 608	Darknet-53	33.0	57.9	34.4	18.3	35.4	41.9

COCO for YOLOv3

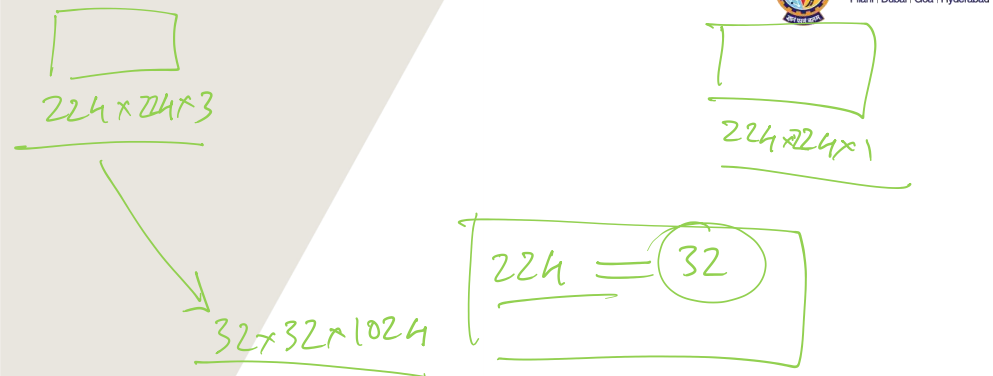
- For clarity, AP@.75 means the AP with IoU=0.75.

COCO mAP

- mAP** (mean average precision) is the average of AP.
- In some context, we compute the AP for each class and average them. But in some context, they mean the same thing. For example, under the COCO context, there is no difference between AP and mAP. Here is the direct quote from COCO:

„AP is averaged over all categories. Traditionally, this is called “mean average precision” (mAP). We make no distinction between AP and mAP (and likewise AR and mAR) and assume the difference is clear from context.“

- In ImageNet, the AUC method is used.
- To summarize: even all of them follow the same principle in measurement AP, the exact calculation may vary according to the datasets. Fortunately, development kits are available in calculating this metric.



Thank you